

Theory 13.8.6

Faisal Memon

April 22, 2025

1 Aggregation by attention

The aggregation methods discussed so far either weight the contribution of the neighbors equally or in a way that depends on the graph topology. Conversely, in graph attention layers, the weights depend on the data at the nodes. A linear transform is applied to the current node embeddings so that:

$$\mathbf{H}'_k = \beta_k \mathbf{1}^T + \Omega_k \mathbf{H}.$$

Then the similarity s_{mn} of each transformed node embedding $\mathbf{h}'_m$ to the transformed node embedding $\mathbf{h}'_n$ is computed by concatenating the pairs, taking a dot product with a column vector ϕ_k of learned parameters, and applying an activation function:

$$s_{mn} = a \left(\phi_k^T \begin{bmatrix} \mathbf{h}'_m \\ \mathbf{h}'_n \end{bmatrix} \right).$$

These variables are stored in a matrix $\mathbf{S}$ of size $N \times N$, where each element represents the similarity of every node to every other. As in dot-product self-attention, the attention weights contributing to each output embedding are normalized to be positive and to one using the softmax operation. However, only those values corresponding to the current node and its neighbors should contribute. The attention weights are applied to the transformed embeddings:

$$\mathbf{H}_{k+1} = \mathbf{a} \left[\mathbf{H}'_k \cdot \mathbf{Softmask}[\mathbf{S}, \mathbf{A} + \mathbf{I}] \right],$$

where $\mathbf{a}[\bullet]$ is a second activation function.

The function $\mathbf{Softmask}[\bullet, \bullet]$ computes the attention values by applying softmax operation separately to each column of its argument $\mathbf{S}$, but only after setting values where the second argument $\mathbf{A} + \mathbf{I}$ is zero to $-\infty$. This ensures that the attention to non-neighboring nodes is zero.

This is very similar to the self-attention computation in transformers (see figure 13.12), except that (i) The keys, queries, and values are all the same, (ii) The measure of similarity is different, and (iii) The attentions are masked so that each node only attends to itself and its neighbors. As in transformers, this system can be extended to use multiple heads that are run in parallel and recombined.